negative frequency, as discussed in Chapter 10 for one-dimensional signals. In equation form, this symmetry is expressed as:

EQUATION 24-2
Symmetry of the two-dimensional frequency domain. These equations can be used in both formats, when the low-frequencies are displayed at the corners, or when shifting places them at the center. In polar form, the magnitude has the same symmetry as the real part, while the phase has the same symmetry as the imaginary part.

$$Re\,X[r,c] = Re\,X[N-r,N-c]$$

$$Im\,X[r,c] = -Im\,X[N-r,N-c]$$

These equations take into account that the frequency spectrum is periodic, repeating itself every N samples with indexes running from 0 to $N-1$. In other words, $X[r,N]$ should be interpreted as $X[r,0]$, $X[N,c]$ as $X[0,c]$, and $X[N,N]$ as $X[0,0]$. This symmetry makes four points in the spectrum match with *themselves*. These points are located at: $[0,0]$, $[0,N/2]$, $[N/2,0]$ and $[N/2,N/2]$.

Each pair of points in the frequency domain corresponds to a sinusoid in the spatial domain. As shown in (a), the value at $[0,0]$ corresponds to the *zero frequency* sinusoid in the spatial domain, i.e., the DC component of the image. There is only one point shown in this figure, because this is one of the points that is its own match. As shown in (b), (c), and (d), other pairs of points correspond to two-dimensional sinusoids that look like waves on the ocean. One-dimensional sinusoids have a *frequency*, *phase*, and *amplitude*. Two dimensional sinusoids also have a *direction*.

The frequency and direction of each sinusoid is determined by the location of the pair of points in the frequency domain. As shown, draw a line from each point to the *zero frequency* location at the outside corner of the quadrant that the point is in, i.e., $[0,0]$, $[0,N/2]$, $[N/2,0]$, or $[N/2,N/2]$ (as indicated by the circles in this figure). The direction of this line determines the direction of the spatial sinusoid, while its length is proportional to the frequency of the wave. This results in the low frequencies being positioned near the corners, and the high frequencies near the center.

When the spectrum is displayed with zero frequency at the center (Fig. 24-9d), the line from each pair of points is drawn to the DC value at the *center* of the image, i.e., $[N/2,N/2]$. This organization is simpler to understand and work with, since all the lines are drawn to the same point. Another advantage of placing zero at the center is that it matches the frequency spectra of *continuous* images. When the spatial domain is continuous, the frequency domain is *aperiodic*. This places zero frequency at the center, with the frequency becoming higher in all directions out to infinity. In general, the frequency spectra of discrete images are displayed with zero frequency at the center whenever people will view the data, in textbooks, journal articles, and algorithm documentation. However, most calculations are carried out with the computer arrays storing data in the other format (low-frequencies at the corners). This is because the FFT has this format.